Synchronous Permanent Magnet Motor

Typ: P62B 355L4 **Rev.1**

Order: 7134295/10 Compiler - Dept.: Mhandrick - V

Code: PTI/PTO / Besecke Date: 19.12.2022

VEM Sachsenwerk GmbH
Pirnaer Landstraße 176
01257 Dresden

1. Rated data - Operation Point (OP1)

power output:	600	[kW]	power factor	-0,94	
voltage:	520	[V]	number of poles	4	
stator current:	726	[A]	number of phases	3	
frequency:	40,00	[Hz]	connection	star	
classification:	Norske Veritas		speed	1200,0	[rpm]
thermal class/rise	H / H		mechanical torque:	4,78	[kNm]
max. altitude:	1000	[m]	duty type	S1	

2. Open-Circuit voltage

min. magnet temp: 38°C / 2200 rpm	U_{KL_max}	965	[V]
max. voltage without chopper and output contactor	$U_d=\sqrt{2} * U_{KL}$	1365	[Vdc]
OP1: magnet temp: 90°C / 1200 rpm	U_{KL}	499	[V]

3. Resistances / Reactances OP 1 (saturated)

	Zn	0,41342	[Ω]		
resistance of statorwinding / phase at 20°C		0,0032	[Ω]	0,772	[%]
direct-axis synchronous reactance	x_d	0,27020	[Ω]	65,36	[%]
quadrature-axis synchronous reactance	x_q	0,28900	[Ω]	69,90	[%]
direct-axis | quadrature-axis sub-transient reactance	x''_d | x''_q	42,50	[%]	56,32	[%]
leakage inductance | reactance	L_σ | x_σ	117	[µH]	7,13	[%]

4. Short-circuit data

max. aperiodic short-circuit current (peak value)	$\hat{I}_s$	4962	[A]		
initial periodic short-circuit current (RMS)	$I''_{k(3)}$	1760	[A]		
continuous periodic short-circuit current (RMS)	I_k	1438	[A]	I_k/I_n	1,980
short-circuit torque	$M''_{k(2)}$	16	[kNm]	$M''_{k(2)}/M_n$	3,303

5. Converter: IVIC Class for Motor: IVIC C | IEC 60034-18-41

allowed rise time:	5	[kV/µs]	allowed rise time:	>0.3	[µs]
allowed voltage peak:	2500	[V]	min. switching frequency:	2500	[Hz]

6. Operating Points

	S [kVA]	U_{KL} [V]	I1 [A]	cosφ rms	P_{el} [kW]	f1 [Hz]	n [rpm]	M_{mech} [kNm]	P_{mech} [kW]	P_v [kW]	η [%]
1	**654**	**520**	**726**	**-0,94**	**612**	**40,00**	**1200**	**4,78**	**600**	**11,5**	**98,1**
2	675	690	564	+0,91	611	73,33	2200	2,61	600	10	98,3

losses to be dissipated P_v: 11 kW + 10% tolerance

7. Cooling system

			IC Cooling code	IC71 W	
max. cooling medium tp.:	38	[°C]	max. % glycol	30	
temperature rise in cw:	3	[K]	pressure drop	< 1	[bar]
water quantity:	70	[l/min]	water quality	freshwater, enclosed loop	

8. General mechanical

Construction	IM B3	Bearing, D-side:	Antifriction bearing, isolated
Type of protection:	IP54	Bearing, N-side:	Antifriction bearing, isolated
Weight:	2.560 +/- 10% [kg]	Earthing brush:	included
Shock resistance:	n. a.	Shaft end:	cylindrical
Vibration level:	IEC 60034-14 Level A	key way:	included

9. Sensors

Winding protection:	2x Pt100 per phase (3 wire)	Leakage sensor:	Baumer Clever Level
Bearing temperature sensor:	2x Pt100 per phase	Heating tapes:	included
Water temperature sensors:	without	Encoder:	Baumer FGHJ 2 HTL or TTL 2048
Bearing vibration monitoring:	prepared for SPM		

All declarations

- with tolerance by DIN EN60034-1
- valid for sinusoidal values only